

OOPS ASSIGNMENT

Question1.

Code :

```
#include<iostream>
```

```
using namespace std;
```

```
class Polygon {
```

```
protected:
```

```
    int width, height;
```

```
public:
```

```
    void set_value(int w, int h) { width = w; height = h; }
```

```
    virtual void calculate_area() = 0;
```

```
};
```

```
class Rectangle : public Polygon {
```

```
public:
```

```
    void calculate_area() {
```

```
        cout << "Area of Rectangle: " << width * height << endl;
```

```
    }
```

```
};
```

```
class Triangle : public Polygon {
```

```
public:
```

```
    void calculate_area() {
```

```
        cout << "Area of Triangle: " << (width * height) / 2 << endl;
```

```
    }
```

```
};
```

```
int main() {  
    Polygon *ptr;  
    Rectangle rect;  
    Triangle tri;  
  
    ptr = &rect;  
    ptr->set_value(10, 5);  
    ptr->calculate_area();  
  
    ptr = &tri;  
    ptr->set_value(10, 5);  
    ptr->calculate_area();  
    return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ tempCodeRunnerFile.CPP -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }  
Area of Rectangle: 50  
Area of Triangle: 25  
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question2.

```
#include<iostream>
```

```
#include<cmath>
```

```
using namespace std;
```

```
class Shape {
```

```
public:
```

```
    virtual void area() = 0;
```

```
    virtual void display() = 0;
```

```
};
```

```
class Circle : public Shape {
```

```
    double radius;
```

public:

Circle(double r) : radius(r) {}

void area() { cout << "Area of Circle: " << M_PI * radius * radius << endl; }

void display() { cout << "Shape: Circle, Radius: " << radius << endl; }

};

class Rectangle : public Shape {

double l, b;

public:

Rectangle(double l, double b) : l(l), b(b) {}

void area() { cout << "Area of Rectangle: " << l * b << endl; }

void display() { cout << "Shape: Rectangle, L=" << l << " B=" << b << endl; }

};

class Triangle : public Shape {

double base, height;

public:

Triangle(double b, double h) : base(b), height(h) {}

void area() { cout << "Area of Triangle: " << 0.5 * base * height << endl; }

void display() { cout << "Shape: Triangle, Base=" << base << " Height=" << height << endl; }

};

int main() {

Shape *s;

Circle c(7); s = &c; s->display(); s->area();

Rectangle r(5,4); s = &r; s->display(); s->area();

Triangle t(6,3); s = &t; s->display(); s->area();

return 0;

}

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Shape: Circle, Radius: 7
Area of Circle: 153.938
Shape: Rectangle, L=5 B=4
Area of Rectangle: 20
Shape: Triangle, Base=6 Height=3
Area of Triangle: 9
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question3.

Code:

```
#include<iostream>
```

```
#include<cmath>
```

```
using namespace std;
```

```
float area(float base, float height) {
    return 0.5 * base * height;
}
```

```
float area(float side) {
    return (sqrt(3.0) / 4.0) * side * side;
}
```

```
float area(float equalSide, float base, int) {
    float h = sqrt(equalSide*equalSide - (base/2)*(base/2));
    return 0.5 * base * h;
}
```

```
int main() {
    cout << "Area of Right Angle Triangle: " << area(6.0f, 4.0f) << endl;

    cout << "Area of Equilateral Triangle: " << area(5.0f) << endl;
    cout << "Area of Isosceles Triangle: " << area(5.0f, 6.0f, 0) << endl;
    return 0;
}
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Area of Right Angle Triangle: 12
Area of Equilateral Triangle: 10.8253
Area of Isosceles Triangle: 12
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question4.

Code:

```
#include<iostream>
```

```
#include<string>
```

```
using namespace std;
```

```
class Student {
```

```
protected:
```

```
    string name;
```

```
    int rollNo;
```

```
public:
```

```
    Student(string n, int r) : name(n), rollNo(r) {}
```

```
    virtual void display() = 0;
```

```
};
```

```
class Engineering : public Student {
```

```
    string branch;
```

```
public:
```

```
    Engineering(string n, int r, string b) : Student(n, r), branch(b) {}
```

```
    void display() {
```

```
        cout << "[Engineering] Name: " << name << ", Roll: " << rollNo
```

```
        << ", Branch: " << branch << endl;
```

```
    }
```

```
};
```

```
class Medicine : public Student {
```

```
    string specialization;

public:

    Medicine(string n, int r, string s) : Student(n, r), specialization(s) {}

    void display() {

        cout << "[Medicine] Name: " << name << ", Roll: " << rollNo

            << ", Spec: " << specialization << endl;

    }

};
```

```
class Science : public Student {

    string subject;

public:

    Science(string n, int r, string s) : Student(n, r), subject(s) {}

    void display() {

        cout << "[Science] Name: " << name << ", Roll: " << rollNo

            << ", Subject: " << subject << endl;

    }

};
```

```
int main() {

    Student *arr[3];

    arr[0] = new Engineering("Alice", 101, "CSE");
    arr[1] = new Medicine("Bob", 202, "Surgery");
    arr[2] = new Science("Carol", 303, "Physics");

    for(int i = 0; i < 3; i++) arr[i]->display();

    return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
[Engineering] Name: Alice, Roll: 101, Branch: CSE
[Medicine] Name: Bob, Roll: 202, Spec: Surgery
[Science] Name: Carol, Roll: 303, Subject: Physics
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question5.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
class Time {
```

```
    int h, m, s;
```

```
public:
```

```
    Time() : h(0), m(0), s(0) {}
```

```
    Time(int h, int m, int s) : h(h), m(m), s(s) {}
```

```
    Time operator+(const Time &t) {
```

```
        int ts = s + t.s;
```

```
        int tm = m + t.m + ts / 60;
```

```
        int th = h + t.h + tm / 60;
```

```
        return Time(th, tm % 60, ts % 60);
```

```
    }
```

```
    void show() {
```

```
        cout << "Time: " << h << "h " << m << "m " << s << "s" << endl;
```

```
    }
```

```
};
```

```
int main() {
```

```
    Time t1(5,15,34), t2(9,53,58), t3;
```

```
    t3 = t1 + t2;
```

```
    t3.show();
```

```
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "C:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Time: 15h 9m 32s
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question6.

Code:

```
#include<iostream>
```

```
#include<cstring>
```

```
using namespace std;
```

```
class STRING {
```

```
    char str[100];
```

```
public:
```

```
    STRING()    { str[0] = '\0'; }
```

```
    STRING(const char *s) { strcpy(str, s); }
```

```
    bool operator==(const STRING &s) {
```

```
        return strcmp(str, s.str) == 0;
```

```
    }
```

```
    STRING operator+(const STRING &s) {
```

```
        STRING res;
```

```
        strcpy(res.str, str);
```

```
        strcat(res.str, s.str);
```

```
        return res;
```

```
    }
```

```
    void display() { cout << str << endl; }
```



```
};
```

```
int main() {  
    STRING s1("Hello"), s2(" World"), s3("Hello");  
  
    STRING s4 = s1 + s2;  
    cout << "Concatenation: "; s4.display();  
  
    if(s1 == s3) cout << "s1 and s3 are Equal" << endl;  
    else      cout << "s1 and s3 are Not Equal" << endl;  
  
    if(s1 == s2) cout << "s1 and s2 are Equal" << endl;  
    else      cout << "s1 and s2 are Not Equal" << endl;  
    return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }  
Concatenation: Hello World  
s1 and s3 are Equal  
s1 and s2 are Not Equal  
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question7.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
class Matrix {
```

```
    int data[2][2];
```

```
public:
```

```
    Matrix() { for(int i=0;i<2;i++) for(int j=0;j<2;j++) data[i][j]=0; }
```

```
    void set(int a,int b,int c,int d) {
```

```
        data[0][0]=a; data[0][1]=b;
```

```
        data[1][0]=c; data[1][1]=d;
    }
    friend Matrix operator*(const Matrix &a, const Matrix &b);
    void display() {
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++) cout << data[i][j] << " ";
            cout << endl;
        }
    }
};
```

```
Matrix operator*(const Matrix &a, const Matrix &b) {
    Matrix res;

    for(int i=0;i<2;i++)
        for(int j=0;j<2;j++)
            for(int k=0;k<2;k++)
                res.data[i][j] += a.data[i][k] * b.data[k][j];

    return res;
}
```

```
int main() {
    Matrix m1, m2, m3;
    m1.set(1,2,3,4);
    m2.set(5,6,7,8);
    m3 = m1 * m2;
    cout << "Result Matrix:" << endl;
    m3.display();
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Result Matrix:
19 22
43 50
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question8.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
class SafeArray {
```

```
    int arr[5];
```

```
    int size;
```

```
public:
```

```
    SafeArray() : size(5) {
```

```
        for(int i=0; i<size; i++) arr[i] = (i+1)*10;
```

```
    }
```

```
    int& operator[](int index) {
```

```
        if(index < 0 || index >= size) {
```

```
            cout << "Error: Index " << index << " out of bounds!" << endl;
```

```
            exit(1);
```

```
        }
```

```
        return arr[index];
```

```
    }
```

```
    void display() {
```

```
        for(int i=0; i<size; i++) cout << "arr[" << i << "] = " << arr[i] << endl;
```

```
    }
```

```
};
```

```
int main() {
```

```
    SafeArray a;
```

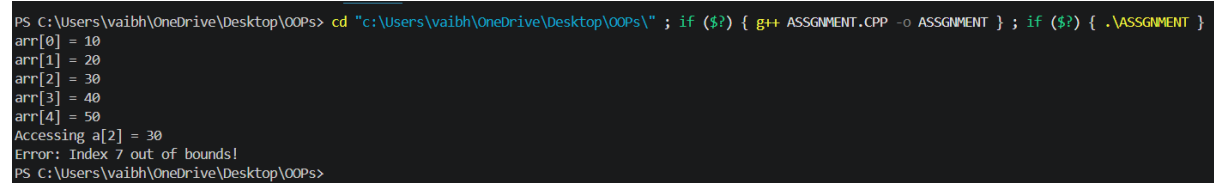
```
    a.display();
```

```
    cout << "Accessing a[2] = " << a[2] << endl;
```

```
    cout << "Accessing a[7] = " << a[7] << endl; // out of bounds

    return 0;
}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
arr[0] = 10
arr[1] = 20
arr[2] = 30
arr[3] = 40
arr[4] = 50
Accessing a[7] = 30
Error: Index 7 out of bounds!
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question9.

Code:

```
#include<iostream>
```

```
using namespace std;
```

```
class Functor {
```

```
    int sum;
```

```
public:
```

```
    Functor() : sum(0) {}
```

```
    // One argument
```

```
    void operator()(int a) {
```

```
        sum = a;
```

```
        cout << "One arg: " << sum << endl;
```

```
    }
```

```
    // Two arguments
```

```
    void operator()(int a, int b) {
```

```
        sum = a + b;
```

```
        cout << "Two args: " << a << " + " << b << " = " << sum << endl;
```

```
    }
```

```
// Three arguments
void operator()(int a, int b, int c) {
    sum = a + b + c;

    cout << "Three args: " << a << " + " << b << " + " << c << " = " << sum << endl;
}
};

int main() {
    Functor f;
    f(10);
    f(10, 20);
    f(10, 20, 30);
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
One arg: 10
Two args: 10 + 20 = 30
Three args: 10 + 20 + 30 = 60
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question10.

Code:

```
#include<iostream>

#include<string>

using namespace std;

class Student {
    string name;

    int age;

    float gpa;
```

```
public:

    friend istream& operator>>(istream &in, Student &s) {

        cout << "Enter Name: "; in >> s.name;

        cout << "Enter Age: "; in >> s.age;

        cout << "Enter GPA: "; in >> s.gpa;

        return in;

    }

    friend ostream& operator<<(ostream &out, const Student &s) {

        out << "Name: " << s.name << ", Age: " << s.age << ", GPA: " << s.gpa;

        return out;

    }

};

int main() {

    Student s;

    cin >> s;

    cout << s << endl;

    return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Enter Name: Vaibhav
Enter Age: 12
Enter GPA: 8
Name: Vaibhav, Age: 12, GPA: 8
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> 
```

Question11.

Code:

```
#include<iostream>

using namespace std;

class Distance {

    float meters;
```

```
public:

    // Conversion constructor: float -> Distance

    Distance(float m) : meters(m) {

        cout << "Constructor called: float -> Distance" << endl;

    }

    Distance() : meters(0) {}

    void display() {

        cout << "Distance: " << meters << " meters" << endl;

    }

};
```

```
int main() {

    float f = 5.75;

    Distance d = f; // implicit conversion

    d.display();

    Distance d2 = Distance(12.5f); // explicit

    d2.display();

    return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Constructor called: float -> Distance
Distance: 5.75 meters
Constructor called: float -> Distance
Distance: 12.5 meters
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> 
```

